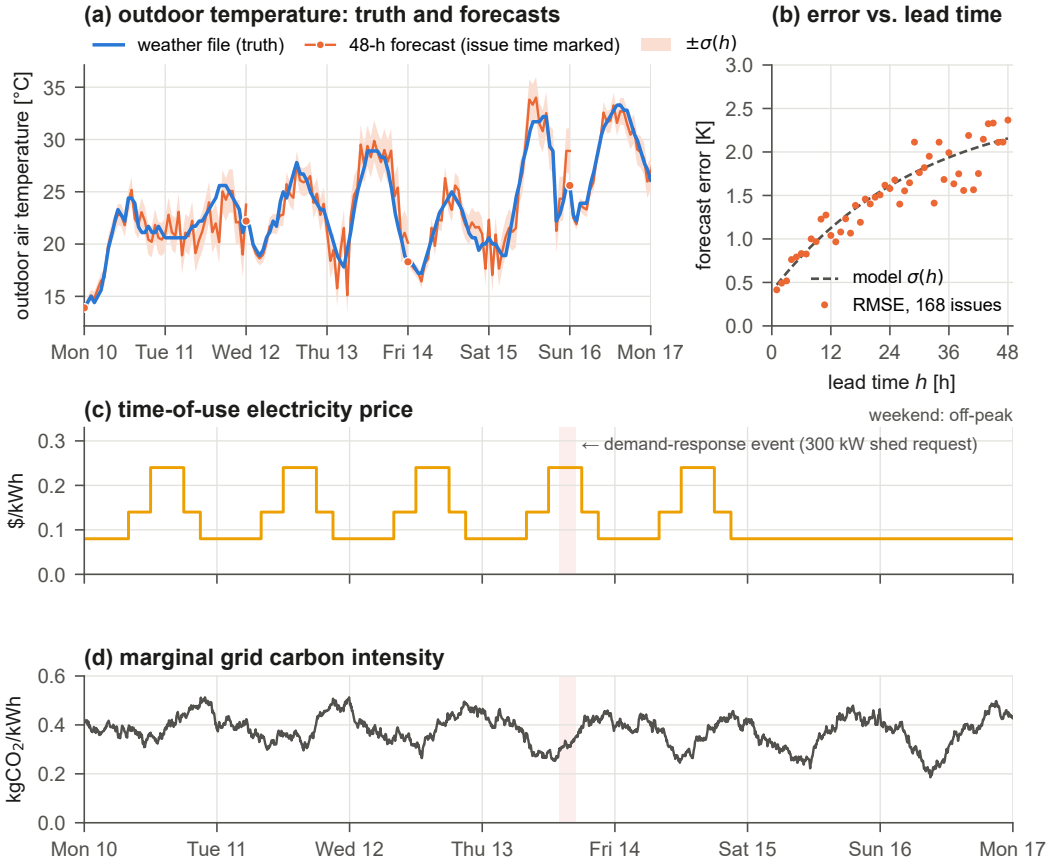

(a) outdoor temperature: truth and forecasts
weather file (truth)
48-h forecast (issue time marked)
±σ(h)
outdoor air temperature [°C]
35
30
25
20
15
Mon 10
Tue 11
Wed 12
Thu 13
Fri 14
Sat 15
Sun 16
Mon 17
(b) error vs. lead time
forecast error [K]
3.0
2.5
2.0
1.5
1.0
0.5
0.0
model σ(h)
RMSE, 168 issues
0
12
24
36
48
lead time h [h]
(c) time-of-use electricity price
weekend: off-peak
← demand-response event (300 kW shed request)
$/kWh
0.3
0.2
0.1
0.0
(d) marginal grid carbon intensity
kgCO2/kWh
0.6
0.4
0.2
0.0
Mon 10
Tue 11
Wed 12
Thu 13
Fri 14
Sat 15
Sun 16
Mon 17